

Delta Basin Energy

Operational Performance & NUPRC Regulatory Compliance (DCOSO/TAR)

A Data-Driven Approach to the Nigerian Upstream Sector

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Project Context: Regulatory Compliance

The Challenge

Under the Petroleum Industry Act (PIA), Nigerian operators face strict scrutiny regarding Technical Allowable Rates (TAR) and the Domestic Crude Supply Obligation (DCOSO).

The Solution

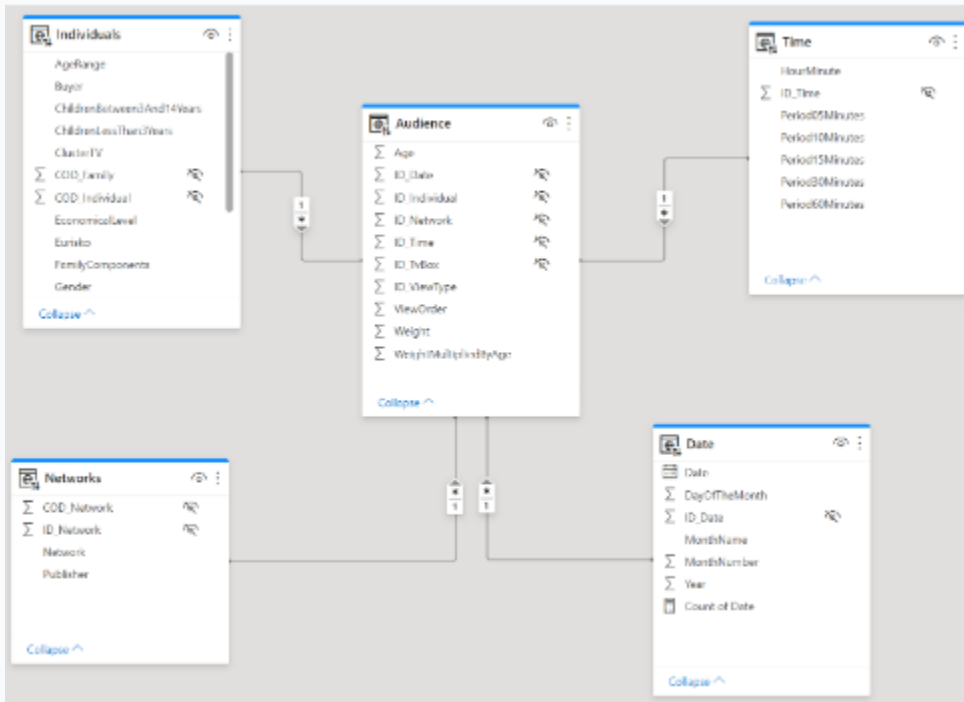
An automated, enterprise-grade Power BI data model designed to track production against regulatory constraints, identifying bottlenecks before they result in penalties.

- Monitor TAR compliance per well string.
- Track DCOSO supply conversion efficiency.
- Analyze flare gas utilization against NGFCP thresholds.



Data Preparation & Modeling

Architecture & DAX



The raw operational dataset was structured into a robust star-schema model to support highly efficient temporal calculations and cross-filtering.



ETL Process

- Cleaned and transformed raw production and supply datasets using Power Query.
- Established a central Date Dimension table for precise chronological trend tracking across 8 quarters.
- Built 1:Many relationships connecting core Fact tables (Production, Gas Utilization) to dimension tables.
- Developed advanced DAX measures for dynamic KPI tracking (e.g., Conversion Trend Analysis, Dynamic Threshold Flags).

Page 1: DCOSO & TAR Compliance Overview



Summary

TAR vs. Actuals Tracking: Automates the comparison between physical production and technical allowable limits, eliminating manual calculation latency across all 8 historical quarters.

Dynamic Constraint Isolation: The lower table dynamically flags underperforming well strings using conditional visual cues, isolating critical drop-offs (e.g., **UZR-01** operating at a **0.75** TAR/MER ratio).

Proactive Intervention: Identifies binding constraints—specifically Sand Cut, High GOR, and BS&W—allowing asset teams to deploy workover maintenance before production deteriorates further.

Page 2: DCOSO Supply Conversion Analysis

The Obligation Gap

Clustered column analysis tracks Volume Allocated vs. Volume Offered vs. Volume Supplied to domestic refineries over an 8-quarter timeline.

Analytical Insight

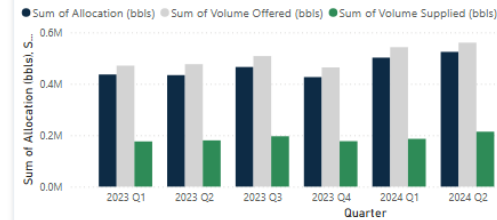
Data proves the company is not short on crude (Offered > Allocation). The failure point is entirely in delivery execution (Supplied < Allocation).

Isolates a major operational bottleneck in Q1 2024, where supply conversion dropped to a series-low of 37.00% right after quarterly regulatory allocations were increased.

Obligation rose faster than delivery capacity—a key flag for supply chain planning.

DCOSO Compliance & Delivery Constraints

Allocation vs. Offered vs. Supplied



Latest Supply Conversion (%)

40.80

Conversion Trend

-3.90

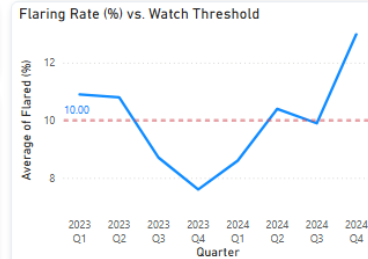
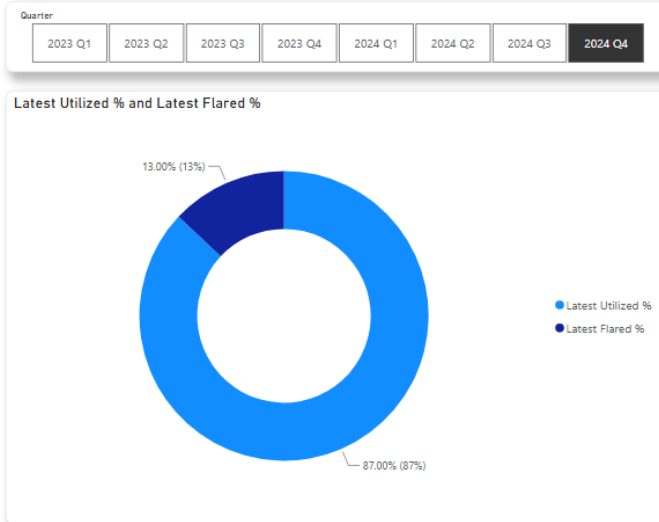
Supply Conversion (%) Trend



2024 Q1 conversion dropped to 37.0% — the lowest point in the series — right after quarterly allocation increased. Obligation rose faster than delivery capacity, a common pattern worth flagging for supply planning

Page 3: Gas Utilization & Flaring Analysis

Delta Basin Energy: Gas Utilization & Flaring Analysis



2024 Q4 flare rate spikes to 13.0% — the highest point in the series, and above the 10% watch threshold. Worth flagging as a flare monetization review candidate

NGFCP Alignment

Monitoring flare rates to ensure compliance with the Nigerian Gas Flare Commercialization Programme.

Watch Thresholds

Implementation of a dynamic 10% Watch Threshold line allows for proactive rather than reactive management.

- **Q4 2024 Spike:** Flare rate spiked to 13.0%, breaching the critical threshold.
- This specific period is flagged immediately as a prime candidate for midstream flare monetization reviews.

Strategic Insights & Stakeholder Recommendations

Key Highlight (The Executive Synthesis) While facility utilization remains strong at 87.00%, systemic bottlenecks—including a 13.00% Q4 flaring spike and a 40.80% DCOSO supply conversion rate—reveal critical gaps between regulatory quotas and physical delivery infrastructure.

Core Insights

- 1 Execution Gap:** DCOSO compliance failures are driven by logistical delivery bottlenecks, not a lack of available crude volume.
- 2 Subsurface Constraints:** The most significant drag on TAR compliance originates from isolated well strings experiencing severe Sand Cut and GOR issues.
- 3 Flaring Volatility:** While overall utilization is strong, intermittent spikes (e.g., Q4 2024) cross the 10% regulatory watch threshold.

Strategic Recommendations

- 1 Supply Chain Audit:** Prioritize an immediate review of domestic delivery logistics to close the gap between Offered and Supplied crude volumes.
- 2 Targeted Intervention:** Direct workover capital specifically toward UZR-01 and ASH-02 to remediate binding Sand Cut constraints and restore TAR ratios.
- 3 Flare Monetization:** Leverage the Q4 2024 flaring data to construct a commercial case for embedded power or mini-LNG under the NGFCP framework.

Stakeholder Recommendations

E&P Operators

1. Prioritize Rig Workovers on "At-Risk" Strings: Deploy maintenance capital directly to underperforming wells (such as UZR-01) to remediate high Sand Cut and water intrusion (BS&W) to lift overall TAR performance above 75.00%.
2. Commercialize Stranded Flare Gas: Treat the 13.00% flaring spike as an investable feedstock rather than an operational waste stream
3. Align Evacuation with Quota Cycles: Bridge the DCOSO conversion gap by improving joint forecasting between subsurface production teams and midstream logistics, ensuring volume offered aligns realistically with pipeline and terminal delivery scheduling.

Regulators

1. Dynamic DCOSO Quota Allocation: Transition from static quarterly allocation targets to dynamic, infrastructure-adjusted quotas. Aligning domestic crude obligations with real-time operator delivery capacity will prevent the severe 37.00% conversion bottlenecks observed during allocation surges.
2. Automated Flaring Surveillance: Move from retrospective penalty enforcement to real-time threshold monitoring. Utilizing automated watch lines (like the 10.00% threshold) allows regulatory bodies to mandate early intervention and gas monetization plans before severe flaring spikes occur.
3. Standardized Constraint Reporting: Mandate uniform digital reporting of binding well constraints (GOR, Sand Cut, BS&W) across all operators to better protect long-term national reservoir health and prevent over-production of mature fields.



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